

Hossein Molavi

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering (Honours, Co-op)

Sept 2021 – Aug 2026 (Expected)

- **Relevant Courses:** Real-Time OS, Microprocessors, Computer Architecture, Systems & Concurrency, DSP

TECHNICAL SKILLS

Languages: C, C++, Python, Bash/Shell, Verilog, ARM Assembly, Java

Hardware/MCUs: STM32, ESP32, nRF, Arduino, ARM Cortex-M, PIC16F887, Raspberry Pi

Protocols: Bluetooth Classic/LE Audio, NFMI, I2C, SPI, UART, CAN, MQTT, USB, TCP/IP

Tools: FreeRTOS, Zephyr, GDB/JTAG/J-Link, CMake/Make, Git, Docker, AWS IoT Core, PlatformIO, KiCad, Linux

Testing: CppUTest, gcov, Pytest, Ellisis BT Analyzer, hardware-in-loop testing

EXPERIENCE

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Waterloo, ON

Wireless Developer

Sept 2025 – Present

Contractor (*Apr 2026 – Present*)

- Architected an autonomous multi-agent AI ecosystem (Orchestrator, Architect, 8 specialized agents) to automate the firmware lifecycle from Jira ticket intake to PR submission.
- Developed a core automation engine leveraging CMake and J-Link for concurrent multi-device firmware deployment, providing the AI agents with real-time, timestamped hardware telemetry.

Intern (*Sept 2025 – Apr 2026*)

- Hardened NFMI driver by resolving DMA race conditions, and built wireless audio interfaces (HFP/A2DP) with mSBC/CVSD codecs.
- Engineered BinauralManager with ACK-based retransmission, and achieved 100% code coverage via 580+ unit tests (CppUTest/gcov).

Asymptote Labs Inc – Anti-Drone Defense Systems

Waterloo, ON

Founder & CTO

Apr 2025 – Present

- Founded defense startup building real-time turret system integrating YOLO detection with motor control
- Designed 4-byte UART protocol with state-machine parsing for low-latency Python-to-Arduino commands
- Engineered converging control system converting pixel offsets to stepper pulses via FOV-calibrated mapping

Wrmth Corp.

North Bay, ON

Embedded Firmware Intern

Jan 2025 – Apr 2025

- Engineered ESP32 C firmware integrating secure BLE/Wi-Fi via MQTT/AWS IoT Core, and architected OTA updates within 512KB RAM.
- Resolved critical memory leaks by debugging race conditions utilizing JTAG and ESP-IDF memory analysis tools.

Midnight Sun Solar Car Team

Waterloo, ON

Embedded Systems Programmer

Aug 2023 – Dec 2023

- Engineered hardware-in-loop test infrastructure and implemented FreeRTOS wrappers for tasks, semaphores, and queues.
- Implemented DMA-driven multi-channel ADC on STM32F1 with FPU acceleration, reducing codesize by 15%.

PROJECTS

Real-Time Operating System | C, ARM Assembly, STM32

- Built preemptive RTOS with EDF scheduling, PendSV/SVC context switching, and 32KB buddy allocator
- Supports 16 concurrent tasks with priority-based preemption and interrupt-driven lifecycle management

RideSafe – Bike Safety System | ESP32-S3, BLE (NimBLE), FreeRTOS, SPI/DMA

- Built dual-MCU cyclist awareness system with BLE communication, radar sensors, LCD, and haptic feedback
- Implemented DMA-enabled SPI display driver at 80MHz with thread-safe FreeRTOS mutex synchronization

NVS Config (Open Source) | C, ESP-IDF

- Published ESP-IDF component providing macro-based type-safe APIs, built-in access control, and auto-save for production firmware.